

Quiz 5

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Please note the quiz is based on the following topics:

- Homomorphisms.
- Isomorphisms.

Problem 1:

Complete the following two definitions. For each, let G and H be two groups, and let $\phi: G \rightarrow H$ be a function.

(a) ϕ is called a(n) _____ if $\phi(xy) = \phi(x)\phi(y)$ _____ $x, y \in G$.

Solution:

ϕ is called a homomorphism if $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$.

(b) ϕ is called a(n) _____ if it is both _____ and _____.

Solution:

ϕ is called an isomorphism if it is both a bijection and a homomorphism.

Problem 2:

Define a homomorphism $\phi: \mathbb{Z}_6 \rightarrow \mathbb{Z}_6$ by $\phi(x) = 3x$. Which elements of $\mathbb{Z}_6$ belong to $\ker(\phi)$?

Solution:

The set of all elements in $\ker(\phi)$ is $\{0, 2, 4\}$.

By the definition of the kernel of a homomorphism, we know that an element $x \in \mathbb{Z}_6$ is in $\ker(\phi)$ if $\phi(x)$ equals the identity element of $\mathbb{Z}_6$. Taking $\mathbb{Z}_6$ to be an additive group, we have that $x \in \ker(\phi)$ if $\phi(x) = 0$. Using the given definition of ϕ and a lil' elbow grease, we can see that because $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$, we have that:

$$\boxed{\phi(0) = (3)(0) \equiv 0 \pmod{6}}$$

$$\phi(3) = (3)(3) \equiv 3 \pmod{6}$$

$$\phi(1) = (3)(1) \equiv 3 \pmod{6}$$

$$\boxed{\phi(3) = (3)(4) \equiv 0 \pmod{6}}$$

$$\boxed{\phi(2) = (3)(2) \equiv 0 \pmod{6}}$$

$$\phi(3) = (3)(5) \equiv 3 \pmod{6}$$

And thus, $\ker(\phi) = \{0, 2, 4\}$.